

Assignment 1: Who's That?

due Monday 06/29 at 9:00am (EST)

After finishing this assignment, submit the `assignment-1.py`, `assignment-1.ipynb`, and `assignment-1-logs.txt` files on Canvas.

Introduction

Welcome to our first homework assignment! Before we begin, let's establish what assignments will look like and what our class expectations are.

In general, our assignments will consist of questions where AI usage is not allowed and those where AI use is expected. Questions where AI use is expected will be explicitly marked as such; if a question has no such label, assume that AI use is not permitted.

You will usually be expected to submit three files on Canvas for every assignment. These are: a `.py` file, where you will write any required code, a `.txt` file with any AI chatbot conversations related to the assignment and any other outside sources you use, and a `.ipynb` file that answers any written questions associated with the assignment.

For this assignment, as with all other work for this course, you are expected to follow the academic integrity guidelines in the course syllabus and Cornell's [Code of Academic Integrity](#).

General Tips + Guidelines

There are several tips and guidelines that you should follow for this assignment:

- You will want to run the code in `assignment-1.py` from the terminal. You can use VSCode's terminal or your computer's terminal app.
- You should run your code in the virtual environment we created in Lab 1. If you have trouble figuring out how to activate it, try Googling! If you're still stuck, reach out to me – the earlier the better.
- Run your code regularly as you write it. It's much easier to catch errors if you debug early and often! You can and should also regularly compare what your program outputs to the expected answers I've included at the end of this document to make sure your code works.
- The assignment is broken down into many tasks and it's easy to skip one by accident. Before submitting, go through each task to make sure you've completed it.
- For this assignment, you shouldn't use, and don't need, any Python constructs not yet covered in class. You should demonstrate your skills with the Python we've learned so far!

Part 0

In this assignment, you'll be extracting and printing information from structured author bios.

Before beginning, make sure that you have downloaded the `assignment-1.pdf` instructions, `assignment-1.py`, and `assignment-1.ipynb` files and the zipped `bios` folder. Within the `info1100` folder you made as part of Lab 1, you should have made an `assignments` subfolder. The files you downloaded should be placed within that subfolder, and the `bios` folder should be unzipped. This will make managing the virtual environment we set up much easier!

All code for this assignment will be written in `assignment-1.py`.

Task 1: Fill in the `# Author` and `# NetID` comments at the top of the `.py` file.

Part 1

You'll notice that the `assignment-1.py` file already has some code in it. You don't need to understand how this code works – we'll learn the content necessary to understand it later in the semester! To summarize, the code will ask you to input the last name of an author. This must be one of the five authors represented in the `bios` folder: “dolan”, “king”, “lahiri”, “morrison”, or “murakami”. It will then load the author bio stored in that `.txt` file and save it as a string in a variable named `bio`. All of the code you are asked to write will only need to reference `bio`.

Currently, when you put in a valid author name the code will print that author's bio. You will replace that print statement with code that fulfills the following tasks. To clarify, the final version of your code should *not* print the entire author biography.

Now, let's get started on extracting author information! To complete the following tasks, you will want a subset of the following String functions:

Method	Description
<code>s.lower()</code>	Returns: a lowercase version of <code>s</code>
<code>s.upper()</code>	Returns: an uppercase version of <code>s</code>
<code>s.title()</code>	Returns: a version of <code>s</code> with all words capitalized
<code>s.count(sub)</code>	Returns: the number of times the substring <code>sub</code> appears in <code>s</code>
<code>s.find(sub)</code>	Returns: the index of the first occurrence of <code>sub</code> in <code>s</code> or <code>-1</code> if <code>sub</code> is not in <code>s</code>
<code>s.strip()</code>	Returns: <code>s</code> with all preceding and trailing whitespace removed
<code>s.replace(old, new)</code>	Returns: <code>s</code> with all instances of the substring <code>old</code> replaced with the substring <code>new</code>
<code>s.islower()</code>	Returns: <code>True</code> if <code>s</code> is lowercase and <code>False</code> otherwise
<code>s.isupper()</code>	Returns: <code>True</code> if <code>s</code> is uppercase and <code>False</code> otherwise
<code>s.istitle()</code>	Returns: <code>True</code> if all words in <code>s</code> are capitalized and <code>False</code> otherwise

You won't need every method in this table, and you may need a few built-in functions that aren't string methods (like `int()` or `round()`). But every *String* method you need is listed above.

As you work through the next tasks, you can double-check your work against the expected answers at the end of this PDF, which shows what the solution code prints for each of the five possible inputs.

Task 2: Extract the author's full name, including any middle names, from `bio` and print it in the form `Name: [name]`.

Task 3: Extract the author's birthday from `bio` and print it in the form `Birthday: [birthday]`. Birthdays are of the form month/day/year and should be printed in that form.

Task 4: Print the day, month, and year of the author's birthday. These print statements should be indented by one tab and preceded by `Day:`, `Month:`, and `Year:` respectively.

Task 5: Extract the author's age in years, as stated in the biography, and print it in the form `Reported Age: [age]`.

Task 6: Calculate the author's age as the number of days between their birthday and the assignment's due date (06/29). For these calculations, you can assume that each month has 30.5 days (the average value). Print the author's age in days in the form `Days Since Birth: [age]`. Their age should be rounded to the nearest integer. Note that there may be multiple valid ways of calculating this; as long as you've done something

reasonable, you will receive points.

Task 7: Extract the total number of siblings the author has and print it in the form `# Siblings: [number of siblings]`. Then, print the number of brothers and sisters the author has with tab indentation and the prefixes `# Sisters:` and `# Brothers:` respectively.

For each of these tasks, make sure that there is no extra whitespace before or after the extracted information.

Part 2 (AI Tasks)

The tasks completed for this part of the assignment all ask you to use an AI chatbot. You may select the AI chatbot of your choice. You should copy and paste every conversation you have with a chatbot related to this assignment into a text file called `assignment-1-logs.txt`. You're expected to understand on a high level any code you include in the assignment; you will be asked to describe all AI-generated code. You may also be asked to explain or edit this code in oral assessments.

For this and Part 3 of the assignment, you will be asked to do some writing! All of written questions should be answered in the `assignment-1.ipynb` file.

Task 8: In the `assignment-1.ipynb` file, fill in the blanks that ask for your name, NetID, and which AI chatbots you use. If you want to use multiple chatbots, specify which one is used for which question!

Task 9: Right now, we are using a numerical value to refer to the month when printing the author's birthday. It'd probably be nicer if we could see the month's name instead! Prompt your chatbot to edit your existing code so that the author's birthday now prints in the format `[month] [day], [year]` and the indented birthday month prints the month's name as well. This code should be pasted below the statement that prints `AI Generated -- Required` in the `assignment-1.py` file.

Task 10: You should continue interacting with the chatbot until you understand the code it generated for Task 9 on a high level (you don't need to understand specific syntax, we'll learn this later). In the `assignment-1.ipynb` file under the Task 10 header write a description – in your own words – of how the Task 9 code works.

Part 3 (AI Free Choice)

Now it's time for you to get creative!

Task 11: Think of something you would like to calculate or extract from the author information included in the biographies! It should be of about the same complexity as the previous tasks in this assignment. Don't restrict yourself to tasks you could write code for independently, the chatbot will help. However, note that there's a balance between getting the chatbot to do cool things and making sure you understand what it's doing and whether

it's actually doing what you want. Once you have an idea, explain it in a few sentences under the Task 11 header in the `assignment-1.ipynb` file.

Task 12: Now, prompt a chatbot to execute the idea you brainstormed in Task 11. Once you have working code, paste it below the statement that prints `AI Generated -- Free Choice` in `assignment-1.py`.

Task 13: You should again prompt the chatbot until you understand broadly what the code you generated for Task 12 does! Under the Task 13 header in the `assignment-1.ipynb` file, describe your Task 12 code in a few sentences.

Task 14: Finally, under the Task 14 header in `assignment-1.ipynb`, reflect in a few sentences on your experience using an AI chatbot to write code for this assignment. What was easy and / or difficult? Did it usually write usable code on your first try, or did you have to iterate some?

Expected Outputs (by Input File)

dolan.txt

```
Author Last Name: dolan

Name: Naoise Dolan
Birthday: 4/14/1992
    Day: 14
    Month: 4
    Year: 1992
Reported Age: 34
Days Since Birth: 13186
# Siblings: 2
    # Sisters: 1
    # Brothers: 1

##### AI Generated - Required #####

Birthday: April 14, 1992
    Day: 14
    Month: April
    Year: 1992

##### AI Generated - Free Choice #####

[YOUR CHOICE HERE]
```

king.txt

Author Last Name: king

Name: Stephen King

Birthday: 9/21/1947

Day: 21

Month: 9

Year: 1947

Reported Age: 78

Days Since Birth: 29452

Siblings: 1

Sisters: 0

Brothers: 1

AI Generated - Required

Birthday: September 21, 1947

Day: 21

Month: September

Year: 1947

AI Generated - Free Choice

[YOUR CHOICE HERE]

lahiri.txt

Author Last Name: lahiri

Name: Jhumpa Lahiri

Birthday: 7/11/1967

Day: 11

Month: 7

Year: 1967

Reported Age: 58

Days Since Birth: 22223

Siblings: 1

Sisters: 1

Brothers: 0

AI Generated - Required

Birthday: July 11, 1967

Day: 11

Month: July

Year: 1967

AI Generated - Free Choice

[YOUR CHOICE HERE]

morrison.txt

Author Last Name: morrison

Name: Toni Morrison

Birthday: 2/18/1931

Day: 18

Month: 2

Year: 1931

Reported Age: 95

Days Since Birth: 35508

Siblings: 3

Sisters: 1

Brothers: 2

AI Generated - Required

Birthday: February 18, 1931

Day: 18

Month: February

Year: 1931

AI Generated - Free Choice

[YOUR CHOICE HERE]

murakami.txt

Author Last Name: murakami

Name: Haruki Murakami

Birthday: 1/12/1949

Day: 12

Month: 1

Year: 1949

Reported Age: 77

Days Since Birth: 28975

Siblings: 0

Sisters: 0

Brothers: 0

AI Generated - Required

Birthday: January 12, 1949

Day: 12

Month: January

Year: 1949

AI Generated - Free Choice

[YOUR CHOICE HERE]